

ORIGINAL ARTICLE

A Smartphone-Based Digital Ruler to Automatically Measure Strabismus in Ophthalmologist-Level: A Prospective, Multicenter Cohort Study

Ruixin Wang , M.D., Ph.D.,¹ Junfeng Lyu , Ph.D.,² Yahan Yang , M.D., Ph.D.,¹ Shaowei Bi , M.Sc., M.D.,¹ Huanyun Yu , M.D., Ph.D.,^{3,4} Hongwei Deng , M.D., Ph.D.,⁵ Mengya Han , M.Sc., M.D.,¹ Weixia Ma , B.Sc.,¹ Minghui Wan , M.Sc., M.D.,^{3,4} Shiming Peng , M.Sc., M.D.,⁵ Xinping Yu , M.D., Ph.D.,¹ Feng Xu , Ph.D.,² and Haotian Lin , M.D., Ph.D.^{1,6,7}

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Abstract

BACKGROUND: Strabismus is a common condition that requires early screening to prevent amblyopia and stereopsis loss. Traditional methods are labor-intensive and costly, limiting their use. A smartphone-based approach for precise, automated strabismus measurement may offer a more accessible solution for early detection and management.

METHODS: We developed the Digital Ruler of Strabismus (DRS), a system that employs three-dimensional reconstruction and deep learning algorithms to analyze 30-second videos of programmatic cover tests recorded by patients using smartphones. The diagnostic and quantitative performance of DRS was evaluated, and its consistency with manual prism measurements performed by ophthalmologists was compared.

RESULTS: A total of 335 participants and 459 videos were collected from three eye centers, with ages ranging from 3 to 72 years. DRS demonstrated a mean absolute error of 4.51 prism diopters (PD) compared with the prism and alternate cover test, indicating good measurement performance. For horizontal deviation, it showed excellent agreement with the prism and alternate cover test, with an intraclass correlation coefficient of 0.98 (95% confidence interval [CI], 0.98 to 0.99). The Bland-Altman analysis further confirmed this agreement, with a mean difference of -1.1 PD and 95% limits of agreement ranging from -11.5 to 9.4 PD, which falls within the clinically acceptable range. With high sensitivity (0.974; 95% CI, 0.949 to 0.987) and specificity (0.938; 95% CI, 0.872 to 0.972), the DRS classification reliably differentiates phoria and tropia from normal ocular alignment. In addition, for intermittent strabismus, DRS can dynamically reconstruct the realignment time and speed during the examination, offering a novel method for observing and measuring the disease dynamics.

CONCLUSIONS: DRS enables strabismus assessment from videos recorded on consumer-grade smartphones. This approach may complement conventional clinic-based examinations and has the potential to expand access to strabismus screening and longitudinal monitoring, particularly in resource-limited settings. (Funded by the National Natural Science Foundation of China; ClinicalTrials.gov number, [NCT05615519](https://clinicaltrials.gov/ct2/show/study/NCT05615519) and others.)

Ruixin Wang, Junfeng Lyu, Yahan Yang, and Shaowei Bi contributed equally to this article.

The author affiliations are listed at the end of the article.

Haotian Lin can be contacted at linht5@mail.sysu.edu.cn or at Zhongshan Ophthalmic Center, Sun Yat-sen University, 7 Jinsui Road, Guangzhou, China, 510623; Feng Xu can be contacted at feng-xu@tsinghua.edu.cn or at the School of Software and BNRist, Tsinghua University, 30 Shuangqing Road, Haidian District, Beijing, China, 100084; and Xinping Yu can be contacted at yuxp@mail.sysu.edu.cn or at Zhongshan Ophthalmic Center, Sun Yat-sen University, 7 Jinsui Road, Guangzhou, China, 510623.

Introduction

Strabismus is a common eye condition that presents during the visual development period in children, and early screening and timely treatment are crucial to prevent lifelong eye misalignment, irreversible amblyopia, and loss of stereopsis.¹⁻⁴ However, early detection remains challenging, as children often lack awareness of visual discomfort, and the subtlety of early signs frequently eludes non-professional caregivers, particularly in phoria or intermittent strabismus cases. Consequently, diagnostic timing varies significantly across individuals,⁵ underscoring the imperative for early screening. From public health and resource optimization standpoints, effective screening should distinguish referral-worthy cases, such as large-angle phorias, intermittent exotropia,⁶ and manifest strabismus, from non-referral cases, including microphorias⁷ and non-strabismic conditions that negligibly affect binocular vision development.

Traditional diagnostic methods, such as cover tests and manual prism, require specialized expertise, are labor-intensive, and incur high costs.⁸ These limitations restrict universal screening of strabismus, especially in primary health care settings. Furthermore, the shortage of pediatric ophthalmologists in many regions exacerbates the problem,^{9,10} making timely and personalized diagnosis more challenging.

In addition, children with strabismus require long-term monitoring and follow-up,⁸ but hospital-based evaluations are typically limited to intervals of 6 months to 1 year, providing only fragmented snapshots of the condition. These periodic assessments may not capture day-to-day fluctuations and can be biased. This is particularly problematic for dynamic conditions^{11,12} such as intermittent exotropia, which is characterized by intermittent alignment or misalignment and an uncertain natural history.^{1,13,14} To assist in clinical decision-making, parents are often asked to report the frequency of eye misalignment observed at home,¹⁵ but these reports are often inaccurate due to parents' lack of clinical training. Furthermore, postoperative drift in ocular alignment is a well-recognized concern. A long-term follow-up study over 21 years reported a recurrence rate of 15.7%,¹⁶ making timely detection of unexpected ocular deviations after surgery essential for prompt intervention. Therefore, there is an urgent need for a novel approach to assist primary health care systems in strabismus screening and timely strabismus progression monitoring. Although automatic measurement methods for strabismus have

emerged,¹⁷⁻²³ their adoption has been limited due to challenges such as insufficient precision and the high cost of specialized equipment.

This study aims to explore the feasibility of achieving precise and automated strabismus measurement using smartphones, addressing the growing need for automation, accuracy, and mobility. Building on our previous development of the first smartphone-based intelligent screening system for visual impairment in infants and young children,^{24,25} we introduce the Digital Ruler of Strabismus (DRS), a method that enables precise and automatic measurement of strabismus deviation from 30-second smartphone-recorded videos. We conducted a prospective multicenter clinical study, comparing DRS with manual prism and alternate cover test (PACT) methods and expert labels.

Methods

STUDY DESIGN AND OVERSIGHT

DRS automatically quantifies the prism diopter (PD) of deviation, identifies the type, and makes a dynamic evaluation of strabismus from a cover-test video (Fig. 1 and Video 1 at ai.nejm.org). A prospective evaluation was carried out at three specialized ophthalmology hospitals from December 2022 to April 2023, with results being compared with expert-level diagnoses and the gold standard of manual prism examinations. The protocol of the study was approved by the Institutional Review Board/Ethics Committee of the Zhongshan Ophthalmic Centre (ZOC) (2023KYPJ101), Sun Yat-sen University, and registered at ClinicalTrials.gov as NCT05615519.

PROCESS STANDARDIZATION AND MEDICAL DEFINITION

To systematize the diagnostic process for integration into computer systems, we designed a standardized examination (Fig. S1 in the Supplementary Appendix). This process included cover-uncover and alternate cover tests performed successively, in accordance with clinical routines. Eye position changes were recorded in videos taken from the coaxial direction. To further deconstruct these clinical videos into computer logic, transforming medical data into a structured format suitable for computational analysis, we subdivided the videos into four segments: fixing with both eyes (segment A); covering-uncovering one eye and the other (segment B); alternating covering one eye and the

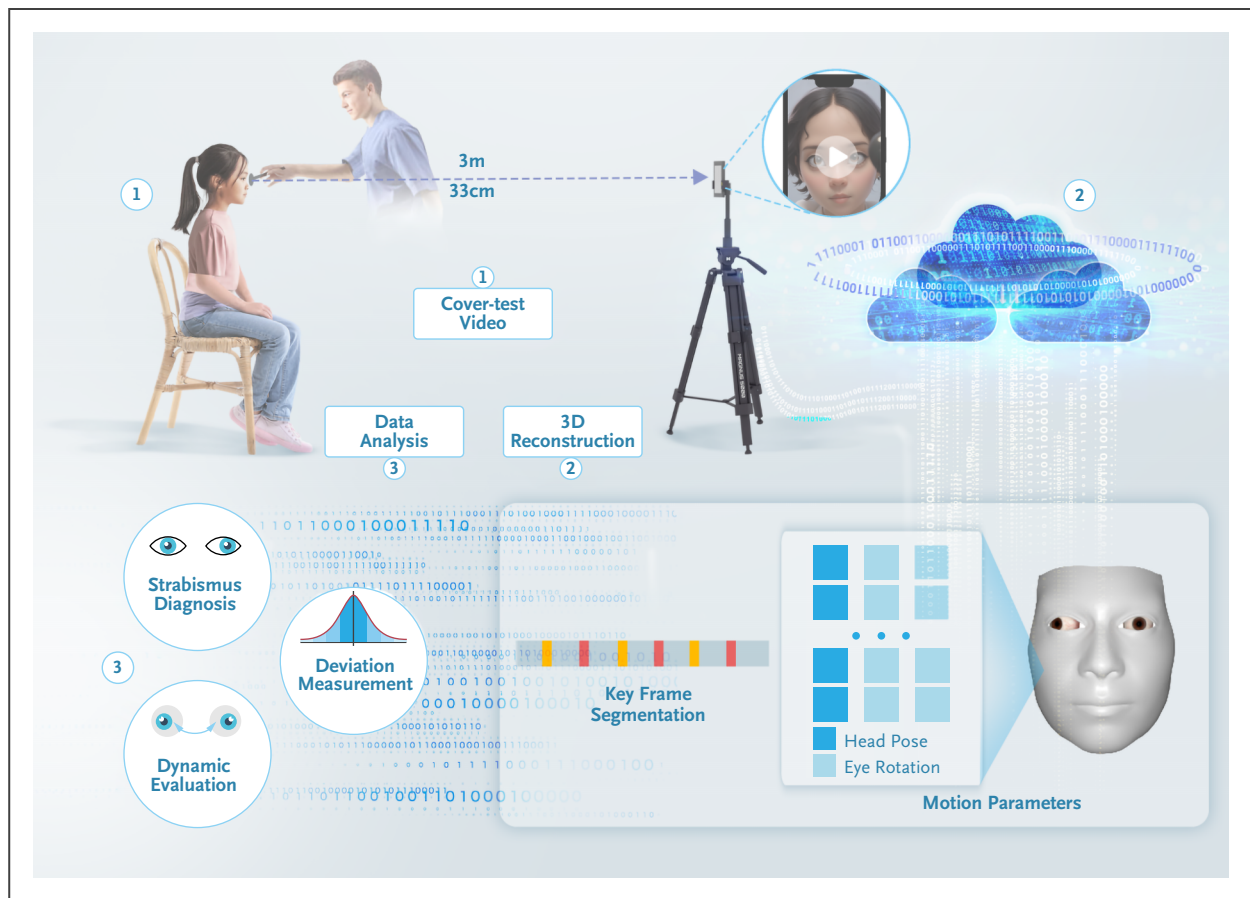


Figure 1. Workflow of the Digital Ruler of Strabismus System.

First, the smartphone-based system records strabismus test videos, including the cover–uncover test and the alternative cover test, where participants look forward at distances of both 3 m and 33 cm. Second, the algorithms utilize 3D reconstruction and deep-learning methods, which could be deployed in the cloud using a supercomputer to analyze the data. Third, the Digital Ruler of Strabismus calculates the motion parameters of the head pose and eye rotations, and extracts key frames to quantify the degree of strabismus and diagnose the condition, as well as support dynamic observation during the examination of intermittent strabismus. 3D denotes three-dimensional.

other (segment C); and fixing with both eyes (segment D). We also introduced the following definitions:

- Reference eye position: The monocular gaze position recorded in segment B. R_0 represents the reference eye position of the right eye, and L_0 represents the reference eye position of the left eye.
- Manifest deviated eye position: The state of deviation when the eyes are fused, recorded in segment A. We calculate the difference between the left and right eyes and their respective reference eye positions, recorded as right eye tropia (R_{tropia}) or left eye tropia (L_{tropia}).
- Maximum deviated eye position: The deviated eye position is induced when fusion is lost through alternate

covering, recorded in segment C. We calculate the difference between the left and right eyes and their respective reference eye positions, recorded as R_{max} and L_{max} .

- Realigned deviated eye position: This is the deviated eye position after alternate covering, recorded in segment D. We calculate the difference between the left and right eyes and their respective reference eye positions, recorded as R_D and L_D .

Strabismus classification definition of DRS:

- Ortho (normal alignment of the eyes)
 - R_{max} and L_{max} less than the threshold

- Tropia (manifest ocular deviation, present with both eyes open and fixating)
 - Rtropia or Ltropia greater than the threshold
- Phoria (latent ocular deviation, compensated under binocular fusion but apparent when fusion is disrupted)
 - Rmax and Lmax greater than the threshold, and RD and LD less than the threshold.

The threshold of 8 PD was applied, following the accepted clinical diagnostic criterion.²⁶ In addition, to explore further applications in mobile medical scenarios, we expanded the intermittent exotropia assessment module to calculate the speed of exotropia recovery in segment D.

- Realigned time: The time for the maximum deviation angle to return to fixating with both eyes without covering.
- Realigned rate: Angular velocity, equal to the ratio of the maximum deviation angle to the realigned time.

Figure S2 abstracts the computer logic of the medical data. Based on the standardized video, the algorithm can analyze different tracking stages by referencing the initial stage for the same subject within the three-dimensional (3D) space.

DEVELOPMENT OF DRS USING 3D MODELING

DRS utilizes 3D reconstruction technology and neural network algorithms to digitize strabismus characteristics, further incorporating medical examination principles (Fig. S1). In brief, our reconstruction method consisted of three main stages: face reconstruction, eyelid reconstruction, and eyeball reconstruction. At each stage, a unique detector was used to extract relevant features (the face landmarks — L_{face} — at the first stage, the eyelid landmarks — L_{eyelid} — and eyelid semantic lines — S_{eyelid} — at the second stage, and the iris landmarks — L_{iris} — at the last stage).

KEY FRAME DETERMINATION

In order to automatically determine the moment of switching covering, we applied segmentation techniques^{27,28} to acquire the masks of covering objects (M_{co}) in each frame. Based on the per-frame M_{co} and L_{eyelid} , we obtained four-class occlusion conditions of each frame (i.e., cover both eyes, cover the left eye, cover the right eye, and cover neither eye). We determined the frame of the maximum deviated eye position by the moment when the occlusion condition changed (e.g., if the occlusion condition changed from “cover the left eye” to “cover the right eye” or “cover neither eye”). The first frame of the new occlusion condition

was taken as the maximum deviated eye position of the left eye. For a single eye, if there were multiple key frames in one video, we took their maximum as the final deviated result.

EYEBALL INITIALIZATION AND TRACKING

DRS further initializes the left and right eyeballs separately by using their own frames of reference for eye position. As shown in Figure S2, given the reference frame of the left eye, we fixed the eyeball rotation and translated the 3D eyeball model to align its iris boundary with that in the image (alignment is based on the iris landmark L_{iris} , as described in Yang et al.²⁵). In the tracking stage, we fixed the eyeball translation and optimized the eyeball rotation to keep its iris boundary aligned with that in the image. For each eye, we optimized its rotation only when it was not occluded, which can be easily achieved by our per-frame occlusion conditions acquired in the key frame determination. To highlight the advantages of the proposed DRS algorithm, which was developed based on 3D reconstruction and neural network modeling, we additionally implemented appearance-based two-dimensional (2D) gaze estimation methods for comparison (Note S1 in the Supplementary Appendix).^{29,30}

PROSPECTIVE CLINICAL ASSESSMENT

To validate the applicability of DRS, we carried out a prospective clinical assessment at Zhongshan Ophthalmic Center, Zhejiang Eye Hospital, and Shenzhen Eye Hospital from December 2022 to April 2023. Inclusion criteria for the study were as follows: participants must be 3 years of age or older, all sexes are eligible, and healthy volunteers are accepted. Vision is not strictly limited, but both eyes must be able to fixate on a target steadily. At the data level, facial videos must meet clinically acceptable quality standards. All videos used in our study met the following minimum technical requirements: frame rate greater than or equal to 25 fps (frames per second) and resolution greater than or equal to 720p minimum dimension (minimum width or height ≥ 720 pixels). The video recording and prism testing were performed in a randomized order, and a rest period of at least 30 minutes was systematically implemented between the two assessments for all participants.

To assist non-specialist users in performing the cover test correctly, we developed a concise, easy-to-understand instructional module that includes video tutorials and on-screen guidance. In addition, the DRS system incorporates real-time quality control using the smartphone’s accelerometer and pixel-based analysis, providing immediate

visual and audio feedback to correct improper positioning or poor video quality (Fig. S3).

During the assessment in each center, two trained non-specialists conducted the cover test and video recording, while an expert also conducted manual prism testing during the same visit. Three certified ophthalmologists with more than 20 years of specialized strabismus practice in tertiary hospitals independently diagnosed ortho, phoria, or tropia for all videos. Final classifications were determined by majority vote. The results obtained from the DRS system were then compared and analyzed alongside the expert's diagnosis and measurements.

STATISTICAL ANALYSIS

To assess the performance of DRS, we evaluated key performance metrics, including sensitivity, specificity, and accuracy for the diagnosis of strabismus. In this three-class classification, ortho was considered normal (negative), whereas tropia and phoria were considered strabismus (positive). The mean absolute error (MAE) was calculated as the average of the absolute differences between the actual DRS and PACT measurements. Intraclass correlation coefficients (ICCs) and Bland-Altman plots were further used to assess agreement between DRS and PACT. ICC values of 0.41 to 0.60, 0.61 to 0.80, and more than or equal to 0.81 were considered indicative of moderate, good, and excellent agreement, respectively.³¹ In addition, scatter plots with fitted linear regression lines were generated to illustrate the agreement between the two methods. Statistical significance was defined as a two-sided P value of less than 0.05. All statistical analyses were performed using R (version 3.3.5) and GraphPad Prism 9.

Results

PATIENT DEMOGRAPHICS AND DATASET

A total of 335 patients and 459 videos met the inclusion criteria, whereas 40 videos of 29 patients were excluded for failure to cooperate with the covering test (Fig. S4). The median age of the patients was 10 years (interquartile range, 6 to 18 years; range, 3 to 72 years). A total of 192 (41.8%) videos were recorded while the participants were wearing glasses. A total of 172 (37.5%) videos recorded distance fixation at 3 m, and 287 (62.5%) at near (33 cm). The median horizontal deviation was -20 PD (range, -80 to +113), and the median vertical deviation was 5 PD (range, 2 to 25). The patient characteristics are listed

Table 1. Characteristics of Participants and Videos.

Characteristic	Value
Median age, years (range)	10 (3 to 72)
Gender, n=cases (%)	
Male	175 (52.2)
Female	160 (47.8)
Glasses, n=videos (%)	
Without	267 (58.2)
With	192 (41.8)
Fixation, n=videos (%)	
Distance	172 (37.5)
Near	287 (62.5)
Strabismus type, n=videos (%)	
Exotropia	261 (46.9)
Esotropia	85 (15.3)
Vertical	211 (37.9)
Median prism, prism diopter (range)	
Horizontal	-20 (-80 to 113)
Vertical	5 (2 to 25)

in [Table 1](#). The validation of the reproducibility of DRS is shown in Figure S5.

MEASUREMENT OF THE DEVIATION OF STRABISMUS

In a dataset of 459 videos displaying horizontal deviation, DRS demonstrated excellent agreement with expert-level PACT assessments. The MAE between the actual DRS and PACT measurements was 4.51 PD, and the ICC reached 0.98 (95% CI, 0.98 to 0.99). Consistent with these findings, the scatterplot with fitted regression line illustrates the close alignment between the two methods ([Fig. 2A](#)). The Bland-Altman plot further confirmed strong agreement, showing a mean difference (bias) of -1.1 PD with 95% limits of agreement (LoA) ranging from -11.5 to 9.4 PD ([Fig. 2B](#)). Among the 211 videos with vertical deviation, DRS also showed agreement with PACT assessments ([Fig. 2C](#)). DRS achieved a MAE of 5.74 PD and demonstrated moderate agreement with an ICC of 0.56 (95% CI, 0.45 to 0.64). The Bland-Altman plot ([Fig. 2D](#)) further indicated a mean difference (bias) of -0.35 PD, with 95% LoA ranging from -15.16 to 14.47 PD.

We assessed DRS in challenging scenarios, specifically in patients with larger deviations and younger in age. In cases with high deviation (≥ 45 PD), DRS demonstrated a MAE of 4.36 PD. The Bland-Altman analysis showed a mean difference of 0.3 PD, with 95% LoA ranging from -10.6 to 10.0 PD (n=26). For children 3 to 5 years of age, DRS achieved a MAE of 4.24 PD. The Bland-Altman analysis showed a

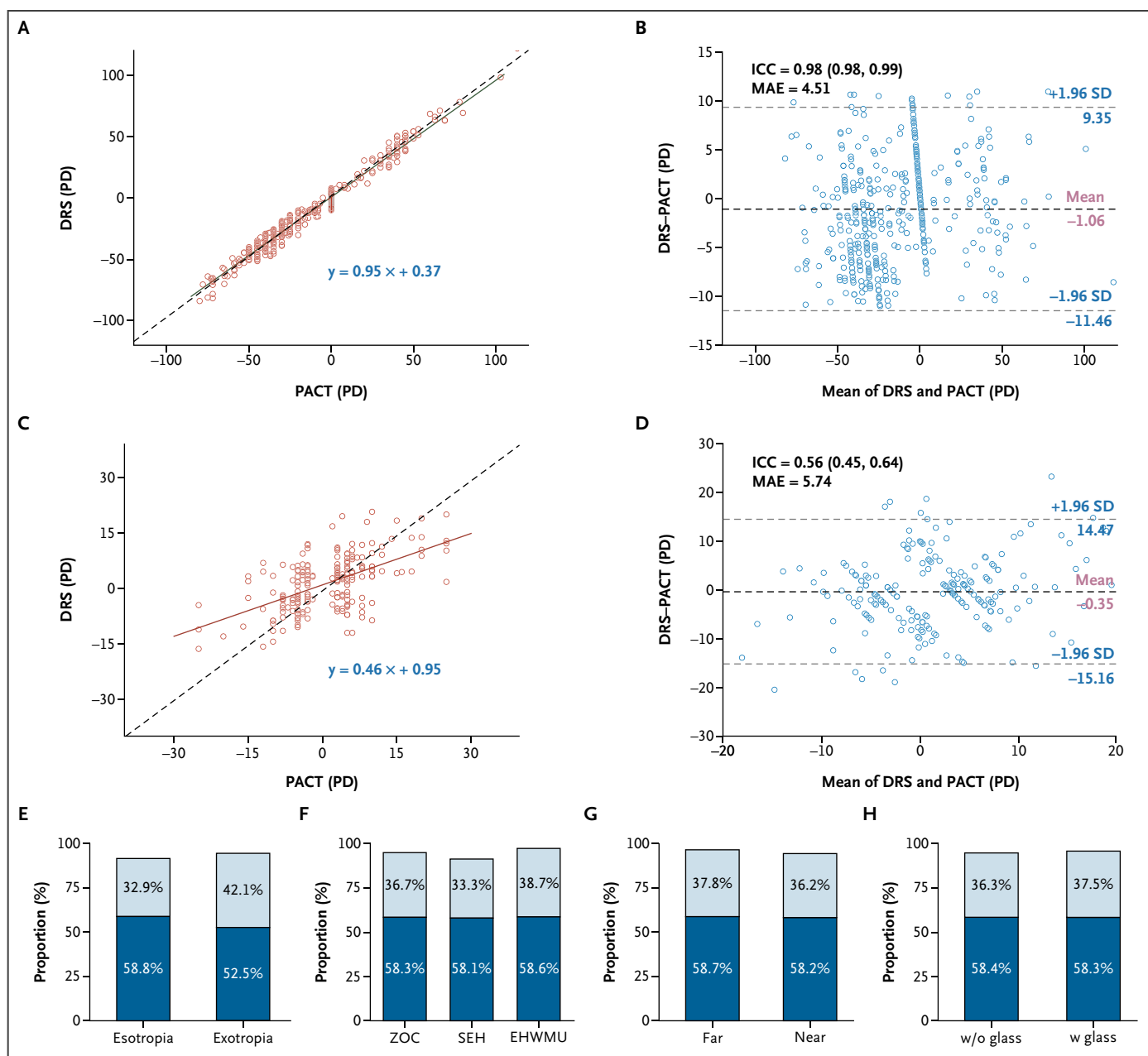


Figure 2. Performance of Quantitative Measurement.

Panels A and B show a comparison of the Digital Ruler of Strabismus (DRS) system and manual prism and alternate cover test (PACT) methods for horizontal strabismus. Panel A shows a scatterplot of DRS and PACT. Panel B shows a Bland–Altman plot. Panels C and D show a comparison of DRS and manual PACT for vertical strabismus. Panel C shows a scatterplot of DRS and PACT. Panel D shows a Bland–Altman plot. Panels E–H show stacked histograms displaying the percentage distribution of measurement errors of DRS (light blue) and manual (dark blue) compared with PACT, categorized by strabismus type, data source, collection distance, and whether or not glasses were worn. Dark blue represents an error of less than 10 prism diopters; light blue represents an error of less than 5 prism diopters. DRS denotes Digital Ruler of Strabismus; EHWMU, Eye Hospital of Wenzhou Medical University; ICC, intraclass correlation coefficient; MAE, mean absolute difference between the actual prism and alternate cover test and Digital Ruler of Strabismus measurements; PACT, prism and alternate cover test; PD, prism diopter; SD, standard deviation; SEH, Shenzhen Eye Hospital; w glass, with glasses; w/o glass, without glasses; and ZOC, Zhongshan Ophthalmic Center.

mean difference of 1.69 PD, with 95% LoA ranging from -11.4 to 8.1 PD (n=79).

An error within 10 PD is generally considered a minor clinical error. DRS showed 95.2% of the errors were within 10 PD, and 58.4% of the errors were within 5 PD, among a total of 459 videos. Stacked histogram plots of error values are depicted in [Figures 2E–2H](#). To validate the robustness of DRS, we analyzed its performance across four subgroups, encompassing three different eye centers, near or far fixation distances, and with or without glasses. The maximum difference between groups was 6.3% for predictions displaying an error of less than or equal to 5 PD (ranging from 52.5% to 58.8%), while the maximum difference between groups was 4.4% for predictions displaying an error of less than or equal to 10 PD (ranging from 91.4% to 95.8%).

DIAGNOSIS OF STRABISMUS OF ORTHO, PHORIA, AND TROPIA

DRS assessed strabismus diagnosis through three key aspects: horizontal strabismus or vertical strabismus; a further classification of ortho, phoria, and tropia; and the direction of deviation (eso or exo). For horizontal strabismus (phoria and tropia as positive; ortho as negative), sensitivity was 0.974 (95% CI, 0.949 to 0.987), specificity was 0.938 (95% CI, 0.872 to 0.972), and accuracy (ACC) was 0.966 (95% CI, 0.943 to 0.979). For vertical strabismus, sensitivity was 0.801 (95% CI, 0.752 to 0.831), specificity was 0.725 (95% CI, 0.559 to 0.849), and ACC was 0.789 (95% CI, 0.748 to 0.825). For eso and exo deviations, sensitivity was 0.999 (95% CI, 0.982 to 0.999), specificity was 0.918 (95% CI, 0.832 to 0.963), and ACC was 0.980 (95% CI, 0.957 to 0.991). The confusion matrices are shown in [Figure 3A and 3B](#) and [Figure S6](#). The analysis of cases that combined horizontal and vertical deviations is shown in [Figure S7](#).

ASSESSMENT OF VARIABILITY OF INTERMITTENT EXOTROPIA

Intermittent exotropia features considerable variability in deviation, making it challenging to objectively assess with existing techniques. Therefore, we have introduced a new temporal dimension within DRS to provide a dynamic evaluation for these patients. [Figure 3C](#) illustrates the characteristic changes in three individual misaligned eyes during refusion; the videos were taken by parents at home. DRS automatically calculated the time required and realignment speed for realignment to provide a basis for assessing the degree of intermittent strabismus and designing surgical interventions. The realignment speeds in the four individual

misaligned eyes were 1.30, 3.03, 2.13, and 0.33 seconds, respectively, while the realignment speeds, defined as the ratio of the maximum deviation angle to the realignment time, were 7.987 degrees/second, 2.344 degrees/second, 6.073 degrees/second, and 70.75 degrees/second.

COMPARISON WITH CURRENT AUTOMATIC METHODS OF STRABISMUS

We compared the performance in strabismus measurements of PACT between DRS and the reported six studies using professional equipment, including eye tracker,^{17,18} video-oculography,^{19,20} and virtual reality,^{21,22} as well as deep-learning diagnostic algorithms based on 2D red, green, blue images of appearance images.³² The mean difference of DRS was 1.1 PD, and the median 95% LoA was 10.4 PD (mean difference). The performance of professional equipment was 1.44 PD plus or minus 11.35 PD (range, 5.1 to 27.9) among the six studies, and the appearance-based method was 2 PD plus or minus 12 PD. Therefore, DRS showed comparable performance to other professional devices ([Fig. 4](#), upper left).^{17–22,32–34} When further comparing the strabismus diagnostic classification capabilities of the reported studies, we found that three previous studies reported qualitative analyses using smartphones,^{32,33} with three reporting sensitivity for binary classification (range, 80.0% to 94.0%) and specificity (range, 76.5% to 99.3%). DRS showed sensitivity of 97.4% and specificity of 93.8%, while the appearance-based method showed sensitivity of 92.1% and specificity of 63.0% ([Fig. 4](#), lower left). In addition, our study included 335 participants, the highest among all studies. DRS achieves balanced functionality and contextual adaptability, enabling quantitative analysis while ensuring mobile compatibility. In addition, DRS adds a dynamic dimension to measurement compared with previous methods. DRS performance was validated against the gold standard, with masking tests incorporated into the study design rather than reliance on Hirschberg or simple convolutional neural network (CNN) methods ([Fig. 4](#), right panel).

Discussion

In a multicenter study of 335 participants, our smartphone-based system, DRS, demonstrated high accuracy and diagnostic performance compared with prism cover testing, supporting its potential role in expanding access to strabismus screening and monitoring.

Several studies have developed automatic methods for strabismus measurement, generally classified as regression- or

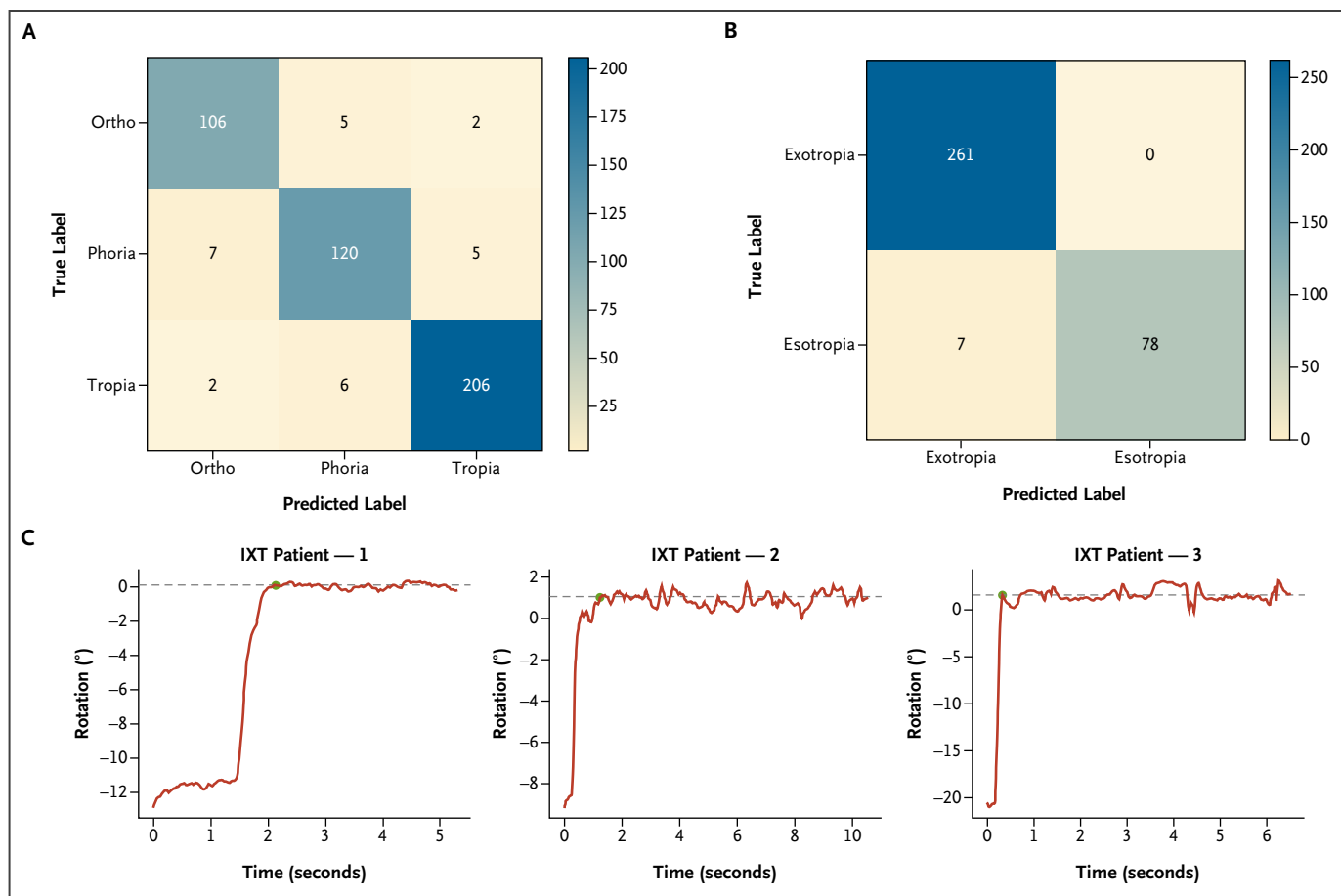


Figure 3. Performance of Strabismus Diagnosis and Dynamic Evaluation.

Panels A and B show a confusion matrix illustrating the classification results of the Digital Ruler of Strabismus. The numbers represent the count of videos in each category, with darker colors indicating larger quantities. Panel A shows three-class results: orthophoria (normal alignment), phoria (latent deviation), and tropia (manifest deviation). Panel B shows two-class results: exotropia (outward eye deviation) and esotropia (inward eye deviation). Panel C shows dynamic measurement of eye position variations in intermittent exotropia. The x-axis represents time, while the y-axis indicates the deviation angle. Red dots indicate the time points at which the deviated eye realigned to the orthotropic position. IXT denotes intermittent exotropia.

classification-based. Regression methods detect geometric features using devices,¹⁷⁻²³ such as eye trackers or infrared cameras, but their high cost, need for calibration, and reliance on the Hirschberg test — limited by kappa angle variability — restrict them to preliminary estimation. Classification methods³² employ red, green, and blue images and CNNs with simpler hardware, but provide only coarse detection without precise quantification. These limitations highlight the need for more accurate and clinically applicable approaches.

The advantage of DRS lies in the interpretability of its algorithm. While prior studies have employed standard CNNs for strabismus classification using only ocular appearance images as inputs, this approach overlooks critical elements

of the medical process. DRS, however, incorporates the standard medical diagnostic logic of cover-uncover and alternating cover tests during its developmental phase. Furthermore, DRS employs a 3D model for gaze estimation. This model visualizes its predicted gaze direction frame by frame. Through landmarks (Video 1), the 3D model renders quantifiable semantic information about iris boundaries and gaze direction, revealing more intermediate computational details than CNN heatmap visualizations, which typically highlight only approximate regions of interest. Although the underlying landmark calculation process is not fully transparent, the current DRS system provides users with interpretable intermediate results.³⁵ Notably, our prior work demonstrated this algorithm's accuracy ($\leq 1\%$ pixel

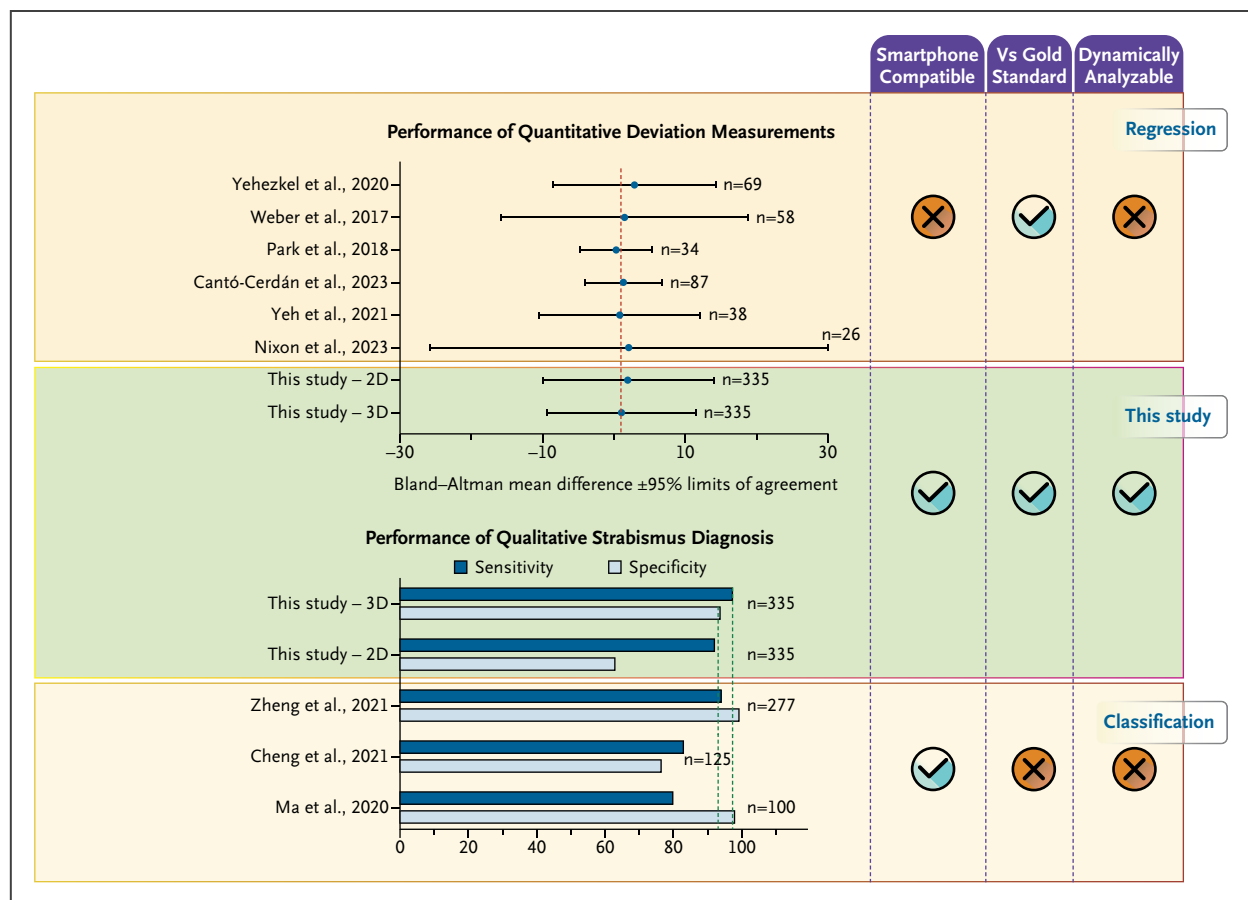


Figure 4. Comparison with Current Automatic Methods of Strabismus.

The forest plot shows the performance of the current automatic strabismus measurement method compared with the traditional gold standard, the prism, and the alternative cover test. The plot reports both the mean difference and the 95% limits of agreement (Bland–Altman analysis), with lower values indicating better performance. The bar chart (lower panel) shows the performance of the proposed strabismus diagnosis methods in distinguishing between healthy individuals and those with strabismus. The chart reports sensitivity and specificity, with higher values indicating better performance. *n* values indicate the number of individuals included in each study. The chart on the right illustrates the performance differences across three functional dimensions. 2D denotes two-dimensional; and 3D, three-dimensional.

error) during development.²⁵ This accuracy persists despite changes in head position or interindividual eyeball difference (Fig. S8).

We have demonstrated that DRS has the ability to diagnose strabismus and measure the degree of deviation. Clinical diagnosis of strabismus is inherently complex, and DRS provides a three-class categorization: orthophoria, phoria, and tropia. It also identifies the direction of deviation (i.e., esotropic, exotropic, or vertical deviations). The level of granularity in classification surpasses previous binary approaches (present or absent), allowing for more detailed categorization. Studies utilizing eye-tracking devices have reported a horizontal strabismus prediction deviation of

plus or minus 11.4 PD (half-width of the 95% LoA).¹⁷ In contrast, DRS achieves a difference deviation of plus or minus 10.4 PD using only a smartphone. Accurate diagnosis and quantification of horizontal strabismus can support tiered referral systems in primary care. For instance, small-angle exophoria (<10 to 15 PD) is common and often requires no intervention if asymptomatic, while unnecessary referrals would strain resources. In contrast, large-angle exophoria and intermittent exotropia, though sometimes cosmetically subtle or asymptomatic, can impair binocular vision development and warrant referral; even small-angle esotropia risks causing amblyopia and requires timely intervention. A key strength of DRS lies in its combination of quantitative and qualitative assessment without specialized

equipment, making it suitable for tiered screening. It may also support resource allocation by reducing overtreatment of low-risk cases and avoiding missed diagnoses of high-risk ones. Furthermore, postoperative patients could use DRS for remote monitoring (with clinician oversight) of residual strabismus, consecutive deviations, or persistent overcorrection.^{36,37}

A foreseeable challenge in real-world application lies in developing adjustment strategies that effectively balance screening or monitoring efficiency, user experience, and detection sensitivity. In addition, DRS's performance in diagnosing vertical strabismus remains suboptimal. Vertical strabismus presents greater clinical complexity, often involving abnormal eye movements and head posture. Its lower prevalence also results in limited training data. Algorithmically, substantial upper eyelid coverage impairs pupil detection accuracy, affecting deviation measurement. To achieve higher precision, future work should integrate measurements of eye movements and other relevant parameters.

In addition to routine measurements of strabismus, DRS can introduce new dimensions for disease assessment over time, providing potential solutions to current clinical challenges. Intermittent exotropia is a common type of strabismus, and there is currently no objective method to assess the degree of the condition, especially when patients transition from an orthotropic to a deviated eye position. Consequently, the variability in the degree of strabismus makes it challenging for doctors to accurately gauge the severity of the condition. Our observations from videos captured by parents highlight the dynamic nature of this condition. In intelligent detection, we have introduced a temporal dimension to objectively measure strabismus variables, such as deviation change speed and recovery time, which traditional methods struggle to provide. Patients with such conditions may undergo home assessments similar to the self-testing conducted by hypertensive patients, presenting a new opportunity for disease diagnosis and evaluation facilitated by DRS.

This work has some limitations. The method may not be applicable to uncooperative individuals during the alternating cover test, such as crying children, those with severe eye tremors, or individuals with significantly impaired monocular vision who are unable to fixate on a target. This requirement for active participation poses a particular challenge for quantitative assessment in very young children, such as infants with infantile esotropia. In addition, individuals with abnormal eye appearances, like

pronounced ptosis, may require manual eyelid manipulation to expose more iris. Further refinement and enhancement of such aspects may be necessary. Our 3D eyeball model is not an exact match to patient eyeballs. The deviation (e.g., eyeball shape, eyeball location, iris radius) may introduce angle errors in the tracking stage. However, this can be improved by personal calibration in future applications.

In summary, we demonstrate that strabismus can be assessed using videos acquired on consumer-grade smartphones. Future work should evaluate its integration into existing clinical workflows and its role in supporting continuity of care for patients with strabismus.

Disclosures

Author disclosures and other supplementary materials are available at ai.nejm.org.

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Author Affiliations

¹Zhongshan Ophthalmic Center, Sun Yat-sen University, WHO Collaborating Centre for Eye Care and Vision, State Key Laboratory of Ophthalmology, Guangdong Provincial Key Laboratory of Ophthalmology and Visual Science, Guangdong Provincial Clinical Research Center for Ocular Diseases, Guangzhou, China

²School of Software and BNRist, Tsinghua University, Beijing, China

³National Clinical Research Center for Ocular Diseases, Eye Hospital, Wenzhou Medical University, Wenzhou, China

⁴School of Ophthalmology and Optometry, Eye Hospital of Wenzhou Medical University, Wenzhou, Zhejiang, China

⁵Shenzhen Eye Hospital, Shenzhen Eye Medical Center, Southern Medical University, Shenzhen, Guangdong, China

⁶Hainan Eye Hospital and Key Laboratory of Ophthalmology, Zhongshan Ophthalmic Center, Sun Yat-sen University, Haikou, Hainan, China

⁷Department of Genetics and Biomedical Informatics, Zhongshan School of Medicine, Institute for Frontier Interdisciplinary Research in Health Sciences and Technology, Sun Yat-sen University, Guangzhou, China

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